A MODULE ADMITTING A LEFT DUAL IN THE MONOIDAL CATEGORY OF MODULES IS FINITE PROJECTIVE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Left dual in a monoidal category, defined by the two duality identities. the author cites Categories, Definition categories-definition-dual at proof lines 18420-18422. Supplies the hypothesis and the identities that the proof manipulates.
- Existence of a surjection from a finite free module onto any module, and the lifting of a map into a tensor product along it. used at proof lines 18434-18440. Produces the auxiliary free module and the lifted unit.
- A module is projective if and only if it is a direct summand of a free module. used at proof lines 18442-18456 to read off the splitting from the commutative diagram. Turns the factorisation found in the diagram into finite projectivity.

Source reference. Original lemma 0FNJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 19569–19569. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 0FFP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 19294–19294. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0FFQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 19295–19295. Navigation pointer to a statement the

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counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let R be a ring. Let M be an R-module. Let N, η, ϵ be a left dual of M in the monoidal category of R-modules, see Categories, Definition 0FFP. Then*

(1) *M and N are finite projective R-modules,*
(2) *the map $e : \mathrm{Hom}_R(M, R) \to N$, $\lambda \mapsto (\lambda \otimes 1)(\eta)$ is an isomorphism,*
(3) *we have $\epsilon(n, m) = e^{-1}(n)(m)$ for $n \in N$ and $m \in M$.*

Proof. The assumptions mean that

$$M \xrightarrow{\eta \otimes 1} M \otimes_R N \otimes_R M \xrightarrow{1 \otimes \epsilon} M \quad \text{and} \quad N \xrightarrow{1 \otimes \eta} N \otimes_R M \otimes_R N \xrightarrow{\epsilon \otimes 1} N$$

are the identity map. We can choose a finite free module F, an R-module map $F \to M$, and a lift $\tilde{\eta} : R \to F \otimes_R N$ of η. We obtain a commutative diagram

$$\begin{array}{ccccc} M & \xrightarrow{\eta \otimes 1} & M \otimes_R N \otimes_R M & \xrightarrow{1 \otimes \epsilon} & M \\ & \searrow^{\tilde{\eta} \otimes 1} & \uparrow & & \uparrow \\ & & F \otimes_R N \otimes_R M & \xrightarrow{1 \otimes \epsilon} & F \end{array}$$

This shows that the identity on M factors through a finite free module and hence M is finite projective. By symmetry we see that N is finite projective. This proves part (1). Part (2) follows from Categories, Lemma 0FFQ and its proof. Part (3) follows from the first equality of the proof. □